



Topic 3 : Federal Flight Agency Network

Group: 1904

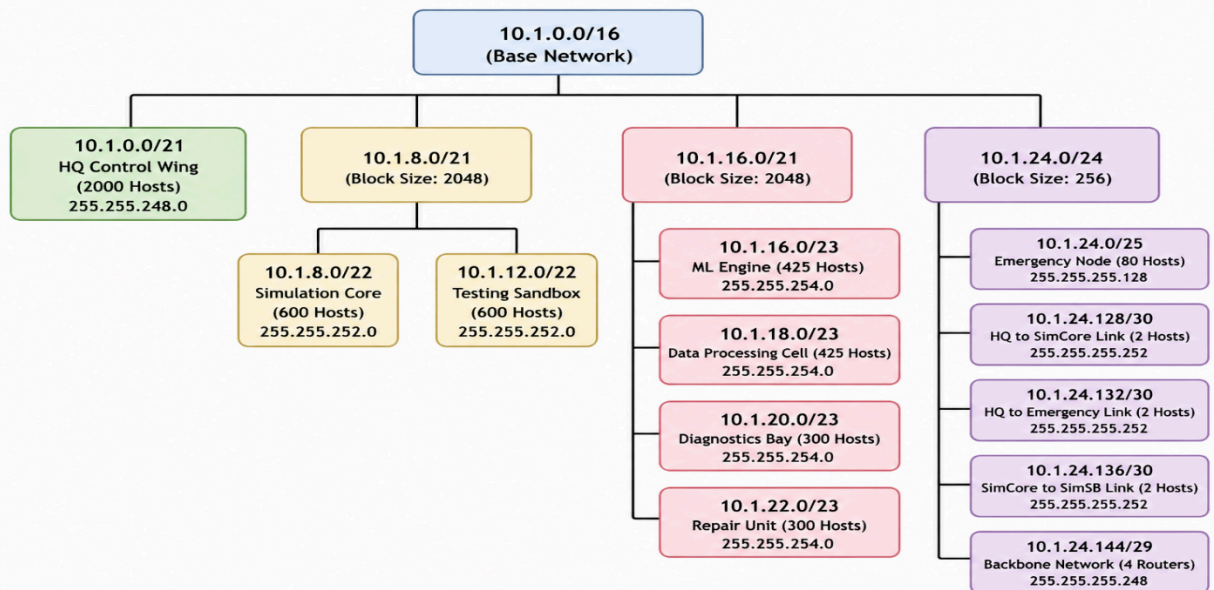
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VLSM Table:

Base Network: **10.1.0.0/16**

Segment	Needed Hosts	Host + 2	Block Size	Host Bits	Prefix
HQ Control Wing	2000	2002	2048	11	/21
Simulation Core	600	602	1024	10	/22
Testing Sandbox	600	602	1024	10	/22
ML Engine	425	427	512	9	/23
Data Processing Cell	425	427	512	9	/23
Diagnostics Bay	300	302	512	9	/23
Repair Unit	300	302	512	9	/23
Emergency Node	80	82	128	7	/25
Backbone	4 routers	6	8	3	/29
HQ ↔ SimCore	2	4	4	2	/30
HQ ↔ Emergency	2	4	4	2	/30
SimCore ↔ SimSB	2	4	4	2	/30

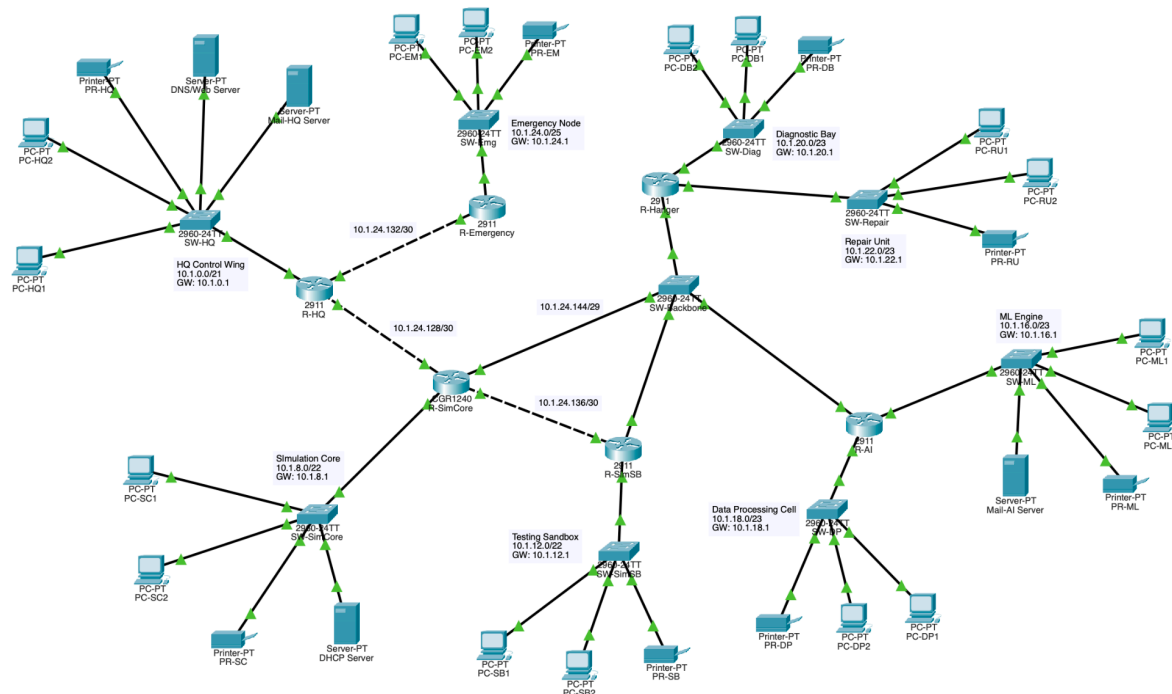
VLSM TREE



IP Address Table:

Segment	Network Address	Prefix	Subnet Mask	Usable Range	Broadcast
HQ Control Wing	10.1.0.0	/21	255.255.248.0	10.1.0.1 – 10.1.7.254	10.1.7.255
Simulation Core	10.1.8.0	/22	255.255.252.0	10.1.8.1 – 10.1.11.254	10.1.11.255
Testing Sandbox	10.1.12.0	/22	255.255.252.0	10.1.12.1 – 10.1.15.254	10.1.15.255
ML Engine	10.1.16.0	/23	255.255.254.0	10.1.16.1 – 10.1.17.254	10.1.17.255
Data Processing Cell	10.1.18.0	/23	255.255.254.0	10.1.18.1 – 10.1.19.254	10.1.19.255
Diagnostics Bay	10.1.20.0	/23	255.255.254.0	10.1.20.1 – 10.1.21.254	10.1.21.255
Repair Unit	10.1.22.0	/23	255.255.254.0	10.1.22.1 – 10.1.23.254	10.1.23.255
Emergency Node	10.1.24.0	/25	255.255.255.128	10.1.24.1 – 10.1.24.126	10.1.24.127
HQ ↔ SimCore	10.1.24.128	/30	255.255.255.252	10.1.24.129 – 10.1.24.130	10.1.24.131
HQ ↔ Emergency	10.1.24.132	/30	255.255.255.252	10.1.24.133 – 10.1.24.134	10.1.24.135
SimCore ↔ SimSB	10.1.24.136	/30	255.255.255.252	10.1.24.137 – 10.1.24.138	10.1.24.139
Backbone	10.1.24.144	/29	255.255.255.248	10.1.24.145 – 10.1.24.150	10.1.24.151

Topology:



Commands for Routers and Servers:

R-HQ:

```
R-HQ>enable
R-HQ#configure terminal

R-HQ(config)#hostname R-HQ

R-HQ(config)#interface GigabitEthernet0/0
R-HQ(config-if)#ip address 10.1.0.1 255.255.248.0
R-HQ(config-if)#no shutdown
R-HQ(config-if)#exit

R-HQ(config)#interface GigabitEthernet0/1
R-HQ(config-if)#ip address 10.1.24.129 255.255.255.252
R-HQ(config-if)#no shutdown
R-HQ(config-if)#exit

R-HQ(config)#interface GigabitEthernet0/2
R-HQ(config-if)#ip address 10.1.24.133 255.255.255.252
R-HQ(config-if)#no shutdown
R-HQ(config-if)#exit

R-HQ(config)#ip route 10.1.24.0 255.255.255.128 10.1.24.134
R-HQ(config)#ip route 10.1.8.0 255.255.252.0 10.1.24.130
R-HQ(config)#ip route 10.1.12.0 255.255.252.0 10.1.24.130
R-HQ(config)#ip route 10.1.16.0 255.255.254.0 10.1.24.130
R-HQ(config)#ip route 10.1.18.0 255.255.254.0 10.1.24.130
R-HQ(config)#ip route 10.1.20.0 255.255.254.0 10.1.24.130
R-HQ(config)#ip route 10.1.22.0 255.255.254.0 10.1.24.130

R-HQ(config)#end
R-HQ#write memory
```

R-Emergency:

```
R-Emergency>enable
R-Emergency#configure terminal

R-Emergency(config)#hostname R-Emergency

R-Emergency(config)#interface GigabitEthernet0/0
R-Emergency(config-if)#ip address 10.1.24.1 255.255.255.128
R-Emergency(config-if)#no shutdown
```

```
R-Emergency(config-if)#exit
```

```
R-Emergency(config)#interface GigabitEthernet0/1  
R-Emergency(config-if)#ip address 10.1.24.134 255.255.255.252  
R-Emergency(config-if)#no shutdown  
R-Emergency(config-if)#exit
```

```
R-Emergency(config)#ip route 10.1.0.0 255.255.248.0 10.1.24.133  
R-Emergency(config)#ip route 10.1.8.0 255.255.252.0 10.1.24.133
```

```
R-Emergency(config)#end  
R-Emergency#write memory
```

R-SimCore:

```
R-SimCore>enable  
R-SimCore#configure terminal
```

```
R-SimCore(config)#hostname R-SimCore
```

```
! ----- VLAN CREATION -----  
R-SimCore(config)#vlan 10  
R-SimCore(config-vlan)#name SIMCORE_LAN  
R-SimCore(config-vlan)#exit
```

```
R-SimCore(config)#vlan 11  
R-SimCore(config-vlan)#name LINK_TO_HQ  
R-SimCore(config-vlan)#exit
```

```
R-SimCore(config)#vlan 12  
R-SimCore(config-vlan)#name LINK_TO_SIMSB  
R-SimCore(config-vlan)#exit
```

```
R-SimCore(config)#vlan 20  
R-SimCore(config-vlan)#name BACKBONE  
R-SimCore(config-vlan)#exit
```

```
! ----- PORT CONFIG -----  
R-SimCore(config)#interface FastEthernet2/3  
R-SimCore(config-if)#switchport mode access  
R-SimCore(config-if)#switchport access vlan 10  
R-SimCore(config-if)#no shutdown  
R-SimCore(config-if)#exit
```

```
R-SimCore(config)#interface GigabitEthernet2/1  
R-SimCore(config-if)#switchport mode access  
R-SimCore(config-if)#switchport access vlan 11  
R-SimCore(config-if)#no shutdown
```

```
R-SimCore(config-if)#exit
```

```
R-SimCore(config)#interface GigabitEthernet2/2
R-SimCore(config-if)#switchport mode access
R-SimCore(config-if)#switchport access vlan 12
R-SimCore(config-if)#no shutdown
R-SimCore(config-if)#exit
```

```
R-SimCore(config)#interface FastEthernet2/4
R-SimCore(config-if)#switchport mode access
R-SimCore(config-if)#switchport access vlan 20
R-SimCore(config-if)#no shutdown
R-SimCore(config-if)#exit
```

```
! ----- SVI (LAYER 3 INTERFACES) -----
```

```
R-SimCore(config)#interface Vlan10
R-SimCore(config-if)#ip address 10.1.8.1 255.255.252.0
R-SimCore(config-if)#ip helper-address 10.1.8.2
R-SimCore(config-if)#no shutdown
R-SimCore(config-if)#exit
```

```
R-SimCore(config)#interface Vlan11
R-SimCore(config-if)#ip address 10.1.24.130 255.255.255.252
R-SimCore(config-if)#no shutdown
R-SimCore(config-if)#exit
```

```
R-SimCore(config)#interface Vlan12
R-SimCore(config-if)#ip address 10.1.24.137 255.255.255.252
R-SimCore(config-if)#no shutdown
R-SimCore(config-if)#exit
```

```
R-SimCore(config)#interface Vlan20
R-SimCore(config-if)#ip address 10.1.24.145 255.255.255.248
R-SimCore(config-if)#no shutdown
R-SimCore(config-if)#exit
```

```
! ----- ROUTING -----
```

```
R-SimCore(config)#ip routing
```

```
R-SimCore(config)#ip route 10.1.0.0 255.255.248.0 10.1.24.129
R-SimCore(config)#ip route 10.1.24.0 255.255.255.128 10.1.24.129
```

```
R-SimCore(config)#router rip
R-SimCore(config-router)#version 2
R-SimCore(config-router)#no auto-summary
R-SimCore(config-router)#network 10.0.0.0
R-SimCore(config-router)#passive-interface Vlan10
R-SimCore(config-router)#passive-interface Vlan11
```

```
R-SimCore(config-router)#exit
```

```
R-SimCore(config)#end
```

```
R-SimCore#write memory
```

R-SimSB:

```
R-SimSB>enable
```

```
R-SimSB#configure terminal
```

```
R-SimSB(config)#hostname R-SimSB
```

```
R-SimSB(config)#interface GigabitEthernet0/0
```

```
R-SimSB(config-if)#ip address 10.1.12.1 255.255.252.0
```

```
R-SimSB(config-if)#ip helper-address 10.1.8.2
```

```
R-SimSB(config-if)#no shutdown
```

```
R-SimSB(config-if)#exit
```

```
R-SimSB(config)#interface GigabitEthernet0/1
```

```
R-SimSB(config-if)#ip address 10.1.24.138 255.255.255.252
```

```
R-SimSB(config-if)#no shutdown
```

```
R-SimSB(config-if)#exit
```

```
R-SimSB(config)#interface GigabitEthernet0/2
```

```
R-SimSB(config-if)#ip address 10.1.24.146 255.255.255.248
```

```
R-SimSB(config-if)#no shutdown
```

```
R-SimSB(config-if)#exit
```

```
R-SimSB(config)#ip route 10.1.0.0 255.255.248.0 10.1.24.137
```

```
R-SimSB(config)#router rip
```

```
R-SimSB(config-router)#version 2
```

```
R-SimSB(config-router)#no auto-summary
```

```
R-SimSB(config-router)#network 10.0.0.0
```

```
R-SimSB(config-router)#passive-interface GigabitEthernet0/0
```

```
R-SimSB(config)#end
```

```
R-SimSB#write memory
```

R-AI (with floating route):

```
R-AI>enable
```

```
R-AI#configure terminal
```

```
R-AI(config)#hostname R-AI
```

```
R-AI(config)#interface GigabitEthernet0/0
```

```
R-AI(config-if)#ip address 10.1.16.1 255.255.254.0
R-AI(config-if)#no shutdown
R-AI(config-if)#exit
```

```
R-AI(config)#interface GigabitEthernet0/1
R-AI(config-if)#ip address 10.1.18.1 255.255.254.0
R-AI(config-if)#no shutdown
R-AI(config-if)#exit
```

```
R-AI(config)#interface GigabitEthernet0/2
R-AI(config-if)#ip address 10.1.24.147 255.255.255.248
R-AI(config-if)#no shutdown
R-AI(config-if)#exit
```

```
R-AI(config)#ip route 10.1.0.0 255.255.248.0 10.1.24.145
R-AI(config)#ip route 10.1.0.0 255.255.248.0 10.1.24.146 69
```

```
R-AI(config)#router rip
R-AI(config-router)#version 2
R-AI(config-router)#no auto-summary
R-AI(config-router)#network 10.0.0.0
```

```
R-AI(config)#end
R-AI#write memory
```

R-Hangar:

```
R-Hangar>enable
R-Hangar#configure terminal
```

```
R-Hangar(config)#hostname R-Hangar
```

```
R-Hangar(config)#interface GigabitEthernet0/0
R-Hangar(config-if)#ip address 10.1.20.1 255.255.254.0
R-Hangar(config-if)#no shutdown
R-Hangar(config-if)#exit
```

```
R-Hangar(config)#interface GigabitEthernet0/1
R-Hangar(config-if)#ip address 10.1.22.1 255.255.254.0
R-Hangar(config-if)#no shutdown
R-Hangar(config-if)#exit
```

```
R-Hangar(config)#interface GigabitEthernet0/2
R-Hangar(config-if)#ip address 10.1.24.148 255.255.255.248
R-Hangar(config-if)#no shutdown
R-Hangar(config-if)#exit
```

```
R-Hangar(config)#ip route 10.1.0.0 255.255.248.0 10.1.24.145
```



```
R-Hangar(config)#router rip
R-Hangar(config-router)#version 2
R-Hangar(config-router)#no auto-summary
R-Hangar(config-router)#network 10.0.0.0
```

```
R-Hangar(config)#end
R-Hangar#write memory
```

DNS/Web Server:

Desktop > IP Configuration

IP Address: 10.1.0.2
Subnet Mask: 255.255.248.0
Default Gateway: 10.1.0.1
DNS Server: 10.1.0.2

Services > DNS

DNS Service: ON

Add DNS Records:

Name: www.federalfgt.control
Address: 10.1.0.2
Click Add

Name: mail.hq.federal
Address: 10.1.0.3
Click Add

Name: mail.ai.federal
Address: 10.1.16.2
Click Add

Name: hqfederal.com
Address: 10.1.0.3
Click Add

Name: aifederal.com
Address: 10.1.16.2
Click Add

Services > HTTP

HTTP: ON
HTTPS: ON

Index.html:

```
<html>  
  <body>  
    <h1>Autonomous Flight Control System Active</h1>  
  </body>  
</html>
```

Mail-HQ Server:

Desktop > IP Configuration

IP Address: 10.1.0.3
Subnet Mask: 255.255.248.0
Default Gateway: 10.1.0.1
DNS Server: 10.1.0.2

Services > EMAIL

SMTP Service: ON
POP3 Service: ON

Domain Name: hqfederal.com
Click Set

User Setup:

Username: hqadmin
Password: 123
Click Add

Mail-AI Server:

Desktop > IP Configuration

IP Address: 10.1.16.2
Subnet Mask: 255.255.254.0
Default Gateway: 10.1.16.1
DNS Server: 10.1.0.2

Services > EMAIL

SMTP Service: ON

POP3 Service: ON

Domain Name: aifederal.com

Click Set

User Setup:

Username: aiadmin

Password: 123

Click Add

DHCP Server:

Desktop > IP Configuration

IP Address: 10.1.8.2

Subnet Mask: 255.255.252.0

Default Gateway: 10.1.8.1

DNS Server: 10.1.0.2

Services > DHCP

Service: ON

Pool Name: serverPool

Default Gateway: 192.168.99.1

DNS Server: 10.1.0.2

Start IP Address: 192.168.99.10

Subnet Mask: 255.255.255.0

Maximum Number of Users: 1

Click Save

Pool Name: SIMCORE_POOL

Default Gateway: 10.1.8.1

DNS Server: 10.1.0.2

Start IP Address: 10.1.8.10

Subnet Mask: 255.255.252.0

Maximum Number of Users: 500

Click Add

Pool Name: TESTING_SANDBOX_POOL

Default Gateway: 10.1.12.1

DNS Server: 10.1.0.2

Start IP Address: 10.1.12.10

Subnet Mask: 255.255.252.0

Maximum Number of Users: 500

Click Add

Full End-to-End Connectivity Validation:

DHCP Validation:

Simulation Core:

PC-SC1 = 10.1.8.11

PC-SC2 = 10.1.8.13

Gateway = 10.1.8.1

DNS = 10.1.0.2

Testing Sandbox:

PC-SB1 = 10.1.12.10

Gateway = 10.1.12.1

DNS = 10.1.0.2

ML Engine:

PC-ML1 = 10.1.16.21

Gateway = 10.1.16.1

DNS = 10.1.0.2

Data Processing Cell:

PC-DP1 = 10.1.18.22

Gateway = 10.1.18.1

DNS = 10.1.0.2

Diagnostics Bay:

PC-DB1 = 10.1.20.21

Gateway = 10.1.20.1

DNS = 10.1.0.2

Repair Unit:

PC-RU1 = 10.1.22.21

Gateway = 10.1.22.1

DNS = 10.1.0.2

R-Hangar DHCP Lease Result:

REPAIR_POOL leased addresses: 3

DIAGNOSTICS_POOL leased addresses: 3

Routing Validation:

Floating Static Route AD Calculation:

First member ID = 22301258

Sum of digits = $2 + 2 + 3 + 0 + 1 + 2 + 5 + 8 = 23$

Number of group members = 3

AD = $23 \times 3 = 69$

R-AI route to HQ:

Primary route:

ip route 10.1.0.0 255.255.248.0 10.1.24.145

Floating backup route:

ip route 10.1.0.0 255.255.248.0 10.1.24.146 69

R-Hangar route to HQ:

Routing entry for 10.1.0.0/21

Known via "static", distance 1

* 10.1.24.145

Default route validation:

show ip route was checked on all routers.

Gateway of last resort was not set on any router.

No default route was used in the network.

Ping and DNS Validation:

PC-HQ1 ping 10.1.0.2 successful

PC-HQ1 ping www.federalftg.control successful

PC-EM1 ping 10.1.0.2 successful

PC-EM1 ping www.federalftg.control successful

PC-SC1 ping 10.1.0.2 successful

PC-SC1 ping www.federalftg.control successful

PC-SB1 ping 10.1.0.2 successful

PC-SB1 ping www.federalftg.control successful

PC-ML1 ping 10.1.24.10 successful

PC-DP1 ping 10.1.0.2 successful

PC-DP1 ping www.federalftg.control successful

PC-RU1 ping 10.1.24.10 successful

PC-ML1 ping 10.1.0.2 successful

PC-ML1 ping www.federalftg.control successful

PC-DB1 ping 10.1.0.2 successful

PC-DB1 ping www.federalftg.control successful

PC-RU1 ping 10.1.0.2 successful

PC-RU1 ping www.federalftg.control successful

Email Validation:

PC-HQ1 sent email to aiuser@aifederal.com successfully.

PC-ML1 received email successfully.

PC-ML1 sent email to hquser@hqfederal.com successfully.

PC-HQ1 received email successfully.

Assumptions Made

- The base network was taken as **10.1.0.0/16** based on the group-size rule.
- Layer 2 switches were assumed to not require management IP addresses.
- Router interface IPs were used as the default gateways for each subnet.
- HQ servers were assigned static IPs so DNS, Web, and Mail services remain reachable.
- DHCP client IPs may vary depending on lease order, but they must stay within the correct subnet.
- Since Packet Tracer does not always allow removing the default DHCP **serverPool**, it was moved to an unused dummy network.
- R-SimCore was configured using VLAN interfaces because its connected ports behaved like switch-module ports.
- Printers in DHCP-based subnets were assumed to receive IPs from their local DHCP pools.

